DA5400W- Foundations of Machine Learning
Tutorial on Classification

QUESTION 1

The Iris dataset contains measurements of iris flowers belonging to three species:

- Setosa
- Versicolor
- Virginica

A subset of observations from this dataset is provided in the table below. Each observation contains two numerical measurements:

- Sepal Length (cm)
- Sepal Width (cm)

These measurements are used as input features for classification.

QUESTION

ID	Sepal Length (cm)	Sepal Width (cm)	Species
1	5.1	3.4	Setosa
2	4.9	3.1	Setosa
3	5.3	3.3	Setosa
4	5.7	2.8	Versicolor
5	5.8	2.7	Versicolor
6	6.0	2.9	Versicolor
7	6.2	3.0	Virginica
8	6.4	2.9	Virginica
9	6.3	3.0	Virginica
10	5.9	3.1	Versicolor

Table 1: IRIS Dataset

QUESTION

A newly observed iris flower has the following measurements:

Feature	Value
Sepal Length	6.0 cm
Sepal Width	3.0 cm

This observation is denoted as the **query point**:

$$x_q = (6.0, 3.0)$$

QUESTION

A newly observed iris flower has the following measurements:

Feature	Value
Sepal Length	6.0 cm
Sepal Width	3.0 cm

This observation is denoted as the **query point**:

$$x_q = (6.0, 3.0)$$

Using the K-Nearest Neighbors algorithm with **k = 3**, determine the **species of the new flower**.

Use **Euclidean distance metric** to compute the distances between the new observation and the training samples.

SOLUTION: STEP 1: Distance Calculation

To compute the similarity between observations, we use the **Euclidean distance**, which is defined as:

$$\sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

where

$x = (x_1, x_2)$ represents the **query point**

$y = (y_1, y_2)$ represents a **training sample**

STEP 1: Distance Calculation

Training Sample 1: $y_1 = (5.1, 3.4)$

Query observation: $x_q = (6.0, 3.0)$

Applying the Euclidean distance formula:

$$d = \sqrt{(6.0 - 5.1)^2 + (3.0 - 3.4)^2}$$

$$d = \sqrt{0.9^2 + 0.4^2}$$

$$d = \sqrt{0.97} = 0.985$$

Thus, the distance between the query point and **Sample 1** is **0.985**.

Similarly, the distances for the remaining samples can be computed using the same formula.

STEP 2: Distance Table

After calculation we have:

ID	Sepal Length (cm)	Sepal Width (cm)	Species	Distance
1	5.1	3.4	Setosa	0.98
2	4.9	3.1	Setosa	1.10
3	5.3	3.3	Setosa	0.76
4	5.7	2.8	Versicolor	0.36
5	5.8	2.7	Versicolor	0.36
6	6.0	2.9	Versicolor	0.10
7	6.2	3.0	Virginica	0.20
8	6.4	2.9	Virginica	0.41
9	6.3	3.0	Virginica	0.30
10	5.9	3.1	Versicolor	0.14

STEP 3: Identify the nearest neighbours (k=3)

For k=3:

ID	Sepal Length (cm)	Sepal Width (cm)	Species	Distance
1	5.1	3.4	Setosa	0.98
2	4.9	3.1	Setosa	1.10
3	5.3	3.3	Setosa	0.76
4	5.7	2.8	Versicolor	0.36
5	5.8	2.7	Versicolor	0.36
6	6.0	2.9	Versicolor	**0.10**
7	6.2	3.0	Virginica	**0.20**
8	6.4	2.9	Virginica	0.41
9	6.3	3.0	Virginica	0.30
10	5.9	3.1	Versicolor	**0.14**

STEP 4: Majority Voting

Among the three nearest neighbors:

- Versicolor → 2
- Virginica → 1

Therefore, using the **KNN algorithm with** $k = 3$, the query observation (6.0 , 3.0) is classified as $\mathbf{Versicolor}$.

QUESTION 2

A dataset records the number of **hours students studied for an exam** and whether they **passed or failed**. The objective is to model the probability that a student will pass the exam based on the number of hours studied.

The data collected from previous students is shown below.

Student	Hours Studied (X)	Result (Y)
1	1	0
2	3	0
3	6	1
4	7	1
5	9	1
6	10	1

QUESTION 2

where

X= number of hours studied

Y=1→ student passed

Y=0→ student failed

Assume the trained model has parameters :

$$\beta_0 = -4$$
$$\beta_1 = 0.8$$

We model the probability of passing using **Logistic Regression** with the following formula:

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$

QUESTION 2

A new student studies for 4 hours before the exam.

1. Compute the probability that this student will pass using the logistic regression model.
2. Using a classification threshold of 0.5, determine whether the student is predicted to pass or fail.
3. Determine the decision boundary, i.e., the number of study hours at which the model predicts a 50% probability of passing.

SOLUTION: STEP 1: Compute the Linear Combination

1. **Probability that this student will pass**

Given,

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$

Let $z = \beta_0 + \beta_1 x$

for $\beta_0 = -4$, $\beta_1 = 0.8$, X = 4

$$z = -4 + 0.8 * 4$$

$$z = -4 + 3.2$$

$$z = -0.8$$

STEP 2: Compute the Probability

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-z}}$$

Substituting the value of z in the equation

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(-0.8)}}$$

$$= \frac{1}{1 + e^{0.8}}$$

$$= \frac{1}{1 + 2.2255}$$

$$P \approx 0.31$$

STEP 3: Classification Decision

So,

$$P(Y = 1 \mid X) \approx 0.31$$

This means there is approximately a 31% chance that the student will pass by studying for 4 hours.

2. Determine whether the student is predicted to pass or fail.

In logistic regression, we convert the predicted probability into a class label using a classification threshold.

Given threshold = 0.5

STEP 3: Classification Decision

Decision rule:

If p >= 0.5 : predict pass (1)
Else if p < 0.5 : predict fail (0)

Since

$$0.31 < 0.5$$

The prediction becomes:

$$Predicted\ Class = 0$$

Therefore, the model predicts that the **student will fail the exam**.

STEP 4: Determine decision boundary

3. **Determine the decision boundary.**

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The decision boundary is the value of x where the model predicts a 50% probability of passing.

This occurs when $P(Y = 1 \mid X) = 0.5$

Using sigmoid function :

$$0.5 = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$

When P=0.5 then $z = \beta_0 + \beta_1 x$ = 0

$$\beta_0 + \beta_1 x = 0$$

Substituting values, we get

STEP 4: Determine decision boundary

$$-4 + 0.8X=0$$

$$0.8X=4$$

$$X = 5$$

Therefore, the decision boundary occurs at 5 study hours.

- If hours studied < 5, probability < 0.5 → predicted Fail
- If hours studied > 5, probability > 0.5 → predicted Pass

QUESTION 3

The following dataset contains 20 animals described using four attributes:

- Give Birth
- Can Fly
- Live in Water
- Have Legs

Each animal is classified as either **Mammal** or **Non-Mammal**.

QUESTION 3

Animal	Give Birth	Can Fly	Live in Water	Have Legs	Class
Human	Yes	No	No	Yes	Mammal
Python	No	No	No	No	Non-Mammal
Salmon	No	No	Yes	No	Non-Mammal
Whale	Yes	No	Yes	No	Mammal
Frog	No	No	Sometimes	Yes	Non-Mammal
Komodo	No	No	No	Yes	Non-Mammal
Bat	Yes	Yes	No	Yes	Mammal
Pigeon	No	Yes	No	Yes	Non-Mammal
Cat	Yes	No	No	Yes	Mammal
Leopard Shark	No	No	Yes	No	Non-Mammal
Turtle	No	No	Sometimes	Yes	Non-Mammal
Penguin	No	No	Sometimes	Yes	Non-Mammal
Porcupine	Yes	No	No	Yes	Mammal
Eel	No	No	Yes	No	Non-Mammal
Salamander	No	No	Sometimes	Yes	Non-Mammal
Gila Monster	No	No	No	Yes	Non-Mammal
Platypus	No	No	No	Yes	Mammal
Owl	No	Yes	No	Yes	Non-Mammal
Dolphin	Yes	No	Yes	No	Mammal
Eagle	No	Yes	No	Yes	Non-Mammal

QUESTION 3

Using the Naïve Bayes classifier, classify the following new animal:

Give Birth	Can Fly	Live in Water	Have Legs
Yes	No	Yes	No

Determine whether the animal is a Mammal or Non-Mammal.

SOLUTION: STEP 1: Compute attribute frequencies

- Mammal(M) - 7
- Non-mammal(N) - 13

Attribute Count

Give Birth

Value	Mammal	Non-Mammal
Yes	6	1
No	1	12

Can Fly

Value	Mammal	Non-Mammal
Yes	1	3
No	6	10

Live in Water

Value	Mammal	Non-Mammal
Yes	2	3
No	3	5
Sometimes	2	5

Have Legs

Value	Mammal	Non-Mammal
Yes	6	10
No	1	3

STEP 2: Baye's Theorem

The posterior probability of a class C given features X is:

$$P(C \mid X) = \frac{P(X \mid C) \cdot P(C)}{P(X)}$$

Where:

$P(C \mid X)$ =probability of class C given attributes X (posterior)

$P(X \mid C)$ =probability of observing attributes X given class C (likelihood)

$P(C)$ =prior probability of class C

$P(X)$ =probability of attributes X (evidence)

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$P(X)$ =probability of attributes X(evidence)

Let the feature vector (attributes) be:

$$X = (A_1, A_2, A_3, A_4)$$

For the animal dataset:

A_1 = Give Birth; A_2 = Can Fly; A_3 = Live in Water; A_4 = Have Legs

STEP 2: Baye's Theorem

Naïve Assumption: Assume all attributes are independent given the class:

$P(X \mid C) = P(A_1, A_2, A_3, A_4 \mid C) = P(A_1 \mid C) \,.\, P(A_2 \mid C) \,.\, P(A_3 \mid C) \,.\, P(A_4 \mid C)$

$$P(C|X) = \frac{P(C) \cdot \Pi_{i=1}^{4} P(A_i|C)}{P(X)}$$

P(X) = P(X | Mammal). P(Mammal) + P(X | Non-mammal). P(Non-mammal)

Since P(X) is the same for all classes, we ignore it for classification:

$$P(C \mid X) \propto P(C) \cdot \prod_{i=1}^{4} P\,(A_i \mid C)$$

This is the Naive Bayes classification rule.

STEP 3: Compute Priors and Conditional Probabilities

We will compute prior P(C) for both classes:

P(C= Mammal) = 7/20 = 0.35

P(C= Non-mammal) = 13/20 = 0.65

Mammals :

Attribute	Value	Probability
Give Birth	Yes	6/7
Can Fly	No	6/7
Live in Water	Yes	2/7
Have Legs	No	1/7

Non-mammals :

Attribute	Value	Probability
Give Birth	Yes	1/13
Can Fly	No	10/13
Live in Water	Yes	3/13
Have Legs	No	3/13

STEP 4: Compute Posterior

Posterior for Mammal:

P(Mammal | X) = P(X | Mammal). P(Mammal) = $\frac{6}{7} . \frac{6}{7} . \frac{2}{7} . \frac{1}{7}$. (0.35)

$$= \frac{72}{2401} . (0.35)$$

= 0.0105

P(Non-mammal | X) = P(X | Non-mammal). P(Non-mammal) = $\frac{1}{13} . \frac{10}{13} . \frac{3}{13} . \frac{3}{13}$. (0.65)

$$= \frac{90}{28561} . (0.65)$$

= 0.00205

STEP 5: Comparing Posterior and Decide

Class	Posterior
Mammal	0.0105
Non-Mammal	0.00205

Since 0.0105 > 0.00205, the animal is classified as : Mammals

QUESTION 4

A hospital has developed a machine learning model to detect breast cancer from mammogram images. The model predicts whether a patient has cancer based on a probability score between 0 and 1. Doctors classify a patient as **Positive (Cancer)** if the probability is ≥ 0.5, otherwise **Negative (Healthy)**.

The model was tested on 200 patients:

- 80 patients actually have cancer
- 120 patients are healthy

After applying the model with a threshold of 0.5, the following results were obtained:

- True Positives (TP): 68 patients correctly predicted as having cancer
- False Positives (FP): 4 healthy patients incorrectly predicted as cancer
- True Negatives (TN): 116 healthy patients correctly predicted as healthy
- False Negatives (FN): 12 cancer patients incorrectly predicted as healthy

QUESTION 4

	Actual Positive	Actual Negative	Total
Predicted Positive	68 (TP)	4 (FP)	72
Predicted Negative	12 (FN)	116 (TN)	128
Total	80	120	

QUESTION 4

a) Compute the accuracy percentage of the model. What does this accuracy indicate about the model's overall performance?

b) Calculate the precision of the model. Interpret the result in the context of medical diagnosis.

c) Calculate the recall (sensitivity) of the model

d) Calculate the F1-score of the model. Explain why the F1-score is useful when both false positives and false negatives are important.

e) Suppose the hospital develops another machine learning model for breast cancer detection with performance as : Precision – 90%, Recall – 92%, Accuracy – 91%

   i. Which model would you recommend for early cancer screening, where the primary goal is to detect as many cancer cases as possible?

   ii. Which model would be more suitable for a confirmatory diagnostic test, where false alarms should be minimized?

   iii. Explain your answer by discussing the trade-off between precision and recall, and how these metrics impact medical decision-making.

SOLUTION: STEP 1: Calculate accuracy

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Accuracy = (Total correct predictions) / (Total number of cases)

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FN+FP}$$

$$\text{Accuracy} = \frac{68+116}{200}$$

$$\text{Accuracy} = \frac{184}{200} = 0.92$$

$$= 92\%$$

The model correctly classifies 92% of the patients, meaning 184 out of 200 patients were predicted correctly as either having cancer or being healthy. This shows that the model performs well overall in making correct predictions. However, accuracy alone is not sufficient in medical diagnosis, because missing cancer cases (false negatives) can have serious consequences, so other metrics like precision and recall are also important.

STEP 2: Calculate precision

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Precision = (Correctly predicted cancer cases) / (Total predicted cancer cases)

Precision = $\frac{TP}{TP+FP}$

Precision = $\frac{68}{72}$ = 0.944

Precision = 94.4%

The precision of the model is 94.4%, meaning that when the model predicts cancer, it is correct 94.4% of the time. This indicates that most patients predicted as having cancer actually have the disease, with only a few false alarms. High precision is important in medical diagnosis because it reduces unnecessary stress and additional medical tests for healthy patients.

STEP 3: Calculate recall

c) Calculate the recall (sensitivity) of the model.

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Recall = (Correctly detected cancer cases) / (Total actual cancer cases)

Recall = $\frac{TP}{TP+FN}$

Recall = $\frac{68}{68+12}$

Recall = $\frac{68}{80}$ = 0.85

Recall = 85%

The recall of the model is 85%, meaning the model correctly detects 85% of actual cancer patients. Out of 80 cancer cases, the model correctly identified 68 patients, while 12 cases were missed. Recall is very important in medical screening because missing a cancer case can delay diagnosis and treatment, so screening systems aim for high recall to detect as many patients as possible.

STEP 4: Calculate F1 Score

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$$F1 = \frac{2 * Precision * Recall}{Precision + Recall}$$

$$F1 = \frac{2 * 0.944 * 0.85}{0.944 + 0.85}$$

$$F1 = \frac{1.6048}{1.794} \approx 0.894$$

$$F1 = 89.4\%$$

The F1-score is 89.4%, indicating that the model maintains a good balance between precision and recall, meaning it both detects many cancer cases and keeps false alarms relatively low.

A value close to 100% shows strong overall performance, so 89.4% suggests the model performs well without heavily sacrificing one metric for the other. This metric is useful in medical diagnosis because it evaluates how effectively the model handles both false positives and false negatives.

STEP 5: Comparing Models

e) Suppose the hospital develops another machine learning model for breast cancer detection with performance as : Precision – 90%, Recall – 92%, Accuracy – 91%

Current Model (Model A) :

- Precision = 94.4%
- Recall = 85%
- Accuracy = 92%

New Model (Model B) :

- Precision = 90%
- Recall = 92%
- Accuracy = 91%

STEP 5: Comparing Models

i. Model for Early Cancer Screening

Model B is recommended for early cancer screening because screening aims to detect as many cancer cases as possible. Model B has a higher recall (92%) than Model A (85%), meaning it identifies more actual cancer patients and misses fewer cases. Therefore, Model B is more suitable for initial screening, even if it may produce slightly more false positives.

ii. Model for Confirmatory Diagnosis

Model A is recommended for confirmatory diagnosis because this stage requires minimizing false positives. Model A has higher precision (94.4%) compared to Model B (90%), meaning its cancer predictions are more reliable. Therefore, Model A is better suited for confirmatory testing, where accuracy of positive results is critical.

STEP 6: Precision - Recall Trade Off

iii. Trade-off Between Precision and Recall

Precision and recall often have a negative correlation.

- Increasing recall usually means detecting more positive cases but may increase false positives.
- Increasing precision reduces false positives but may cause the model to miss some true cases.

In medical decision-making:

- High recall is preferred for screening, because the goal is to detect as many potential cancer cases as possible.
- High precision is preferred for diagnosis, because doctors want to be confident that positive results truly indicate the presence of cancer.

Therefore:

- **Model B is better for screening**
- **Model A is better for confirmatory diagnosis**

Quick Question :

What will happen if we increase or decrease the threshold value?

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Increase threshold: Suppose to 0.8

The model becomes stricter in predicting positive. False positives decrease and precision increases, but more true cases may be missed, so recall decreases.

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Increase threshold: Suppose to 0.8
The model becomes stricter in predicting positive. False positives decrease and precision increases, but more true cases may be missed, so recall decreases.

Decrease threshold: Suppose to 0.3
The model becomes less strict and predicts more positives. Recall increases because more true cases are detected, but false positives increase, so precision decreases.